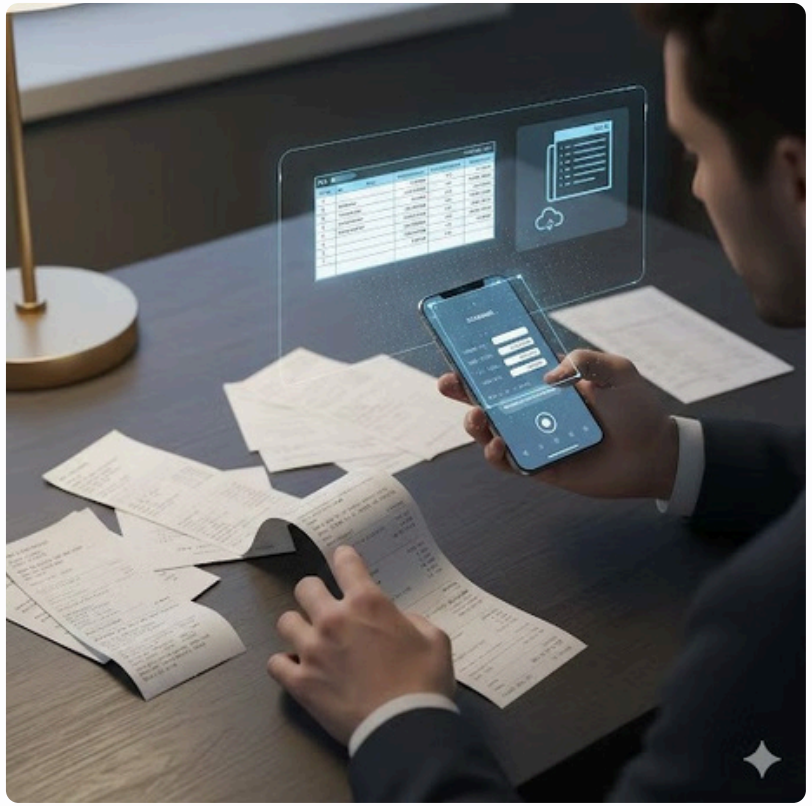
Receipt & Invoice Digitizer

An AI-Powered Automated Expense & Document Management System that streamlines financial record-keeping.



Presented By
Tisha Gurjar

Problem Statement

Manual handling of financial documents is inefficient and problematic.

Manual Handling Challenges

- Time-consuming
- Error-prone
- Loss of financial records
- Difficult expense tracking
- Repetitive data entry

Project Overview

This project builds a system that automatically scans, extracts, and digitizes information from receipts and invoices using OCR (Optical Character Recognition) and NLP-based field extraction. The digitized data is stored in a structured format, making it easy to search, analyze, and integrate with accounting or ERP systems.

1. Automated OCR-based text extraction from scanned receipts and invoices.
2. Field-level parsing for date, vendor, amount, tax, and line items.
3. Web-based dashboard for uploading files, reviewing extracted data, and downloading CSV/Excel reports.
4. Error handling & validation (e.g., duplicate receipts, invalid totals).

Implementation Roadmap

Our project delivery will follow a structured, phased approach over 8 weeks, ensuring each module is robustly developed and integrated.

Milestone 1: Weeks 1-2

Document Ingestion & OCR

- Implement file upload.
- Preprocess images for OCR.
- Extract raw text from sample receipts.

Milestone 2: Weeks 3-4

Field Extraction & Validation

- Apply regex + NLP to parse key fields.
- Validate totals and detect duplicates.
- Store structured results in DB.

Milestone 3: Weeks 5-6

Dashboard & Reporting

- Build Streamlit dashboard for upload/review.
- Add CSV/Excel export.
- Display simple analytics (monthly totals).

Milestone 4: Weeks 7-8

Polishing & Integration

- Improve extraction accuracy with template-based parsing.
- Add search/filter in dashboard.
- Optimize DB queries and reports.

Module 1: Core Functionality Objectives

Our initial development phase focuses on establishing the fundamental capabilities required for efficient document digitization and data extraction.

File Upload & Ingestion

Enabling users to easily upload receipt and invoice images or PDFs for automated processing within the system.

Image Preprocessing for OCR

Implementing algorithms to enhance the quality of uploaded documents, maximizing the accuracy of subsequent text recognition.

Optical Character Recognition (OCR)

Developing robust OCR capabilities to precisely extract raw textual data from all scanned receipts and invoices for further analysis.

Core Functionality Workflow

This detailed workflow illustrates the essential steps from initial document ingestion to final management, highlighting the automated processes and AI integration at each stage.

- API Key Validation
- Document Upload
- Document Preprocessing
- OCR Text Extraction
- Text Cleaning & Normalization
- Duplicate Detection
- AI-Based Data Parsing
- Database Storage
- Receipt Viewing & Management

User Interface & Application Control Module

This module acts as the central entry point and controller, ensuring a seamless and intuitive user experience by managing application flow and user interactions.

Application Layout

Configures the **Streamlit** application layout for a clean, organized, and responsive user interface

API Key & Access

Handles Gemini API key(**Gemini 2.5 Flash**) input, validation, and restricts access until a valid key is provided.

Tab Organization

Organizes the application into **Upload, Dashboard, and History** tabs for clear workflow separation.

Receipt Display

Displays stored receipts and detailed line items through **interactive tables**.

Module Coordination

Controls data flow and interaction between **OCR, AI parsing, and database** modules

Core Features

Uses **session state** for API verification, persistent sidebar settings, and interactive selections.

This module is paramount in controlling the overall flow and user experience of the Digitizer application.

Document Ingestion & OCR Processing Module

This module serves as the initial gateway for all documents, transforming raw inputs into structured, machine-readable text.

Document Upload & Conversion

Supports multiple file formats (JPG, JPEG, PNG, PDF) and converts multi-page PDFs into individual images for processing..

Image Preprocessing

Applies critical enhancements like resizing, denoising, and thresholding to optimize image quality for OCR accuracy.

Tesseract OCR Extraction

Extracts raw textual content using Tesseract OCR, optimized for financial documents such as receipts and invoices.

Text Cleaning & Normalization

Cleans and standardizes OCR output to remove noise and prepare text for AI-based parsing.

Ultimately, this foundational module converts physical or digital documents into usable, structured text, ready for intelligent data extraction.

AI-Based Data Extraction & Validation Module

This module is the intelligence layer of the project, leveraging advanced AI to transform unstructured OCR text into structured, reliable financial data for seamless integration and analysis.

AI-Powered Data Extraction

- Sends cleaned OCR text to Gemini AI using strict JSON prompting for structured output.

Key Field Identification

- Extracts essential fields including vendor name, date, total amount, tax, and line items.

Rigorous Data Validation

- Verifies extracted data through numeric checks, mandatory field validation, and line-item consistency.

Quality Filtering

- Discards incomplete or low-confidence extractions to maintain data reliability.

Deterministic AI Output

- Ensures consistent results using deterministic AI settings (temperature = 0) with JSON recovery logic.

Field Normalization

- Standardizes extracted values into consistent formats for accurate storage and processing.

Data Storage, Duplicate Detection & Management Module

This module ensures data integrity, persistence, and efficient retrieval, forming the backbone of the Digitizer's reliable record-keeping.

Database Management

- Initializes and manages the **SQLite database** for robust operations.
- Securely stores both **receipt-level and line-item data**.
- Efficiently retrieves stored receipts and items for display across the application.

Data Integrity & Quality

- Performs **early duplicate detection** using OCR text to prevent redundant processing.
- Maintains **strong relational integrity** between receipts and their individual line items.

Robust Storage Features

- Utilizes a **normalized database schema** for optimal structure and efficiency.
- Ensures **secure and persistent local storage** of all financial documents.

Ultimately, this module guarantees reliable and duplicate-free data storage, critical for accurate financial record management.

Thank You